Gregorio Ferreira

Computational Mechanical Engineer | Physics-Based FEA | Solid Mechanics | R&D Virtual Prototyping

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Professional Summary

Computational Mechanical Engineer (PhD) with 5+ years of combined applied research and industrial R&D experience in physics-based simulation, solid mechanics, structural dynamics, and non-linear finite element analysis (FEA). Specialized in Abaqus-based simulation, custom solver routines (UMAT/VUMAT/UEL), Python automation, and experimental model validation (DIC, optical microscopy, mechanical testing). Experienced in delivering simulation-driven design and structural optimization across composite structures, hydrogen pressure vessels, tooling, and robotic manufacturing cells.

Core Technical Skills

Computational FEA: Non-linear FEA; solid mechanics; structural dynamics; buckling & stability; parametric analysis; design verification.

Solver Extension: Abaqus Standard/Explicit; custom subroutines (UMAT/VUMAT/UEL); Python automation; MATLAB; FORTRAN.

Materials & Failure: Continuum damage mechanics; progressive failure; multiscale RVE modeling; constitutive modeling; defect correlation.

Verification & Validation: Model V&V; test–simulation correlation; full-field strain (DIC); optical microscopy; mechanical testing; dimensional inspection.

Engineering R&D: Virtual prototyping; structural optimization; CAD/SolidWorks; DoE; technical documentation; cross-functional collaboration.

Applications: Composite pressure vessels; robotic manufacturing cells; multi-material assemblies; load-bearing structures.

Professional Experience

Bernal Institute – University of Limerick
Sep 2022 – Present

Postdoctoral Researcher — Limerick, Ireland

- Develop physics-based non-linear FEA models for structural stiffness, stability, and failure analysis to guide design decisions and verify performance prior to manufacturing.
- Automate simulation workflows in Python–Abaqus, linking parametric model generation, batch execution, and automated post-processing.
- Validate numerical models against physical test data using mechanical testing, full-field strain measurement (DIC), and optical microscopy image analysis.
- Develop an open-source design-to-manufacture pipeline integrating trajectory generation, FE-based structural sizing, and automated robot code generation.
- Apply computational models to physical systems including pressure vessels, tooling, and robotic cells, verifying kinematic reachability and clearance via off-line virtual cell simulation.

HyFloatComp — Hydrogen Pressure Vessels (Simulation / R&D) — Apr 2025 – Present

- Conduct parametric structural analysis to optimize geometry and laminate layup of composite pressure vessels prior to committing to physical tooling.

VariComp — Variable-Stiffness Composite Structures — Sep 2022 – Mar 2025

- Built FE buckling and strength models for tailored structural panels and quantified defect sensitivity using microscale RVE modeling informed by optical microscopy.

HyCo – Hybrid Composites Technology
Jun 2021 – Aug 2022

R&D Mechanical Engineer — São Paulo, Brazil

- Led simulation-driven redesign of load-bearing thermoplastic and metal assemblies using 3D CAD, FEA, and topology optimization, achieving a 50% weight reduction with 15% improved structural performance.
- Carried products from initial concept to full production release through virtual prototyping, physical tooling, mechanical testing, dimensional inspection, and supplier qualification within an ISO 9001 environment.
- Translated engineering requirements into practical design specifications and technical documentation while collaborating across cross-functional R&D teams.

Education

University of São Paulo
2014 – 2019

PhD in Mechanical Engineering — Aeronautical Structures
São Paulo, Brazil

- **Thesis:** *Unified Finite Element Formulations for Composite Plate Structures.*
- Developed custom Abaqus User Element (UEL) and UMAT subroutines to resolve through-thickness stress profiles and progressive damage under implicit integration, validating model predictions against physical test campaigns.

University of São Paulo
2012 – 2014

MSc in Mechanical Engineering — Aeronautical Structures
São Paulo, Brazil

- Implemented a progressive damage model for explicit dynamic FEA as an Abaqus VUMAT subroutine, simulating transient impact response and validating failure evolution against experimental data.

Selected Applied Research Projects

Multiscale Modelling of Composite Material Structures
2018 – 2020

University of São Paulo
São Paulo, Brazil

- Developed multiscale damage models by scripting automated FEA workflows and custom material routines.
- Extended Abaqus implicit capabilities to couple higher-order element formulations with progressive damage, validating results against tensile, bending, and impact test campaigns.

Damage Prediction and Monitoring in Composite Structures
2015 – 2019

University of São Paulo
São Paulo, Brazil

- Applied continuum damage mechanics to predict dynamic failure and residual strength under impact.
- Correlated FE vibration models against measured natural frequencies and mode shapes using laser vibrometry.

Selected Publications & Technical Output

Journal Papers (Peer-Reviewed)

- **Ferreira, G.F.O.**, et al. "From virtual to actual assisted tape placement — application of the Frenet frame to robotic steering trajectories." *Composites Part A*, 2024. [doi:10.1016/j.compositesa.2024.108369]
- **Ferreira, G.F.O.**, et al. "A finite element unified formulation for composite laminates in bending considering progressive damage." *Thin-Walled Structures*, 2022. [doi:10.1016/j.tws.2021.108864]
- **Ferreira, G.F.O.**, et al. "Development of a finite element via Unified Formulation: implementation as a User Element subroutine to predict stress profiles in composite plates." *Thin-Walled Structures*, 2020. [doi:10.1016/j.tws.2020.107107]

International Conference Proceedings

- **Ferreira, G.F.O.**, et al. "Continuous Trajectory Generation for Automated Manufacturing of Super-Ellipsoidal Composite Pressure Vessels." *ACM7*, Delft, 2026.
- **Ferreira, G.F.O.**, et al. "A Virtual Environment Based on the Frenet Frame for Manufacturing Variable Angle Tow Composite Structures with LATP." *ECCOMAS*, Lisbon, 2024.
- **Ferreira, G.F.O.**, et al. "Exploring Void Defects in LATP Manufacturing through 3D Microscale Modeling." *ICCS27*, Ravenna, 2024.

Peer reviewer for *Composite Structures* (Elsevier). Five peer-reviewed journal papers and international conference presentations.

Additional Information

- **Languages:** English (fluent); Portuguese (native).
- **Work Authorisation:** Full Irish work authorisation (Stamp 4); open to relocation to the Netherlands.
- **References:** Available on request.